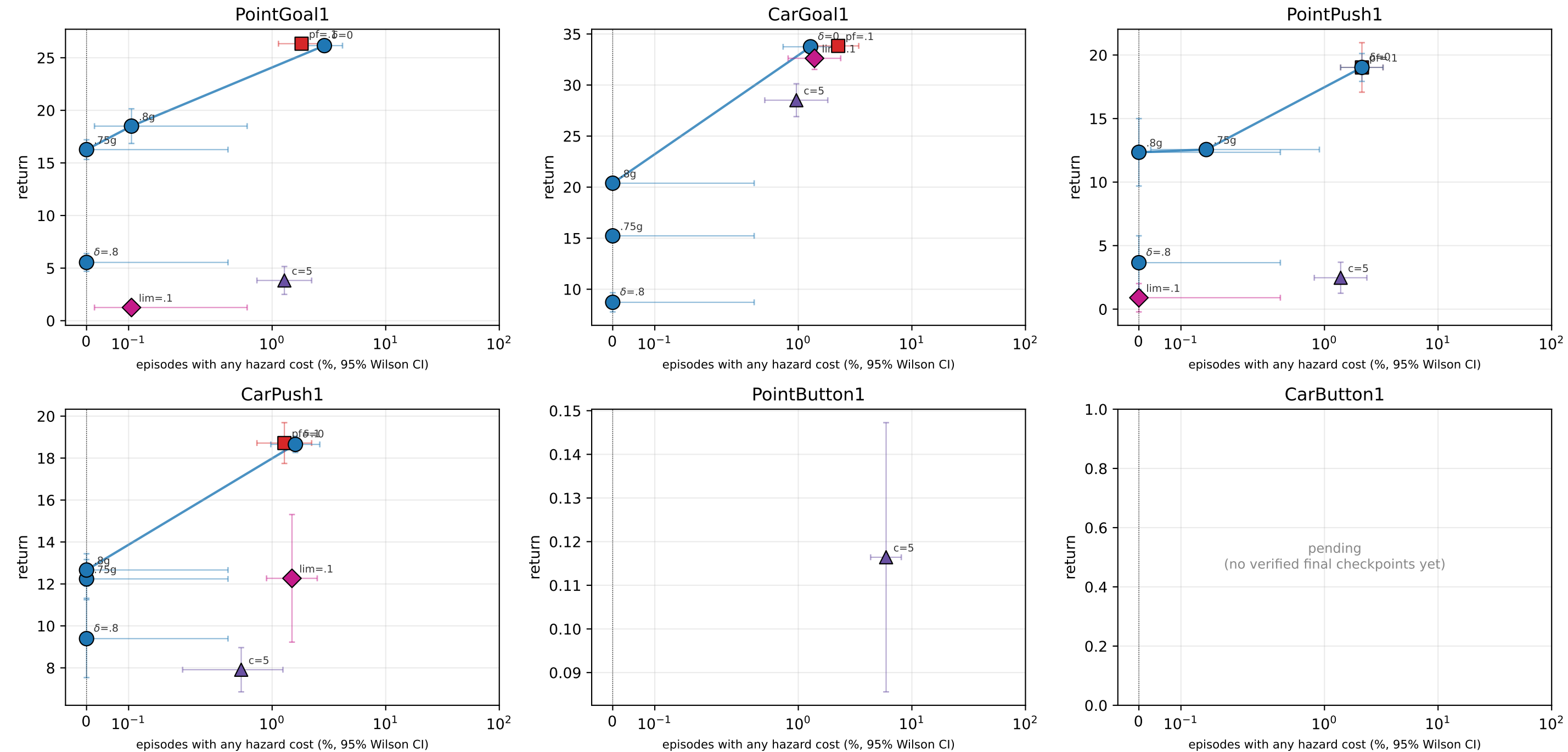


Return vs. violation rate (final checkpoint, 312 episodes/seed; single cost channel; violation = any cost in the episode)



● SafeOT-typed (ours)
 ▲ SAC + fixed penalty (penalty-based)
 — typed Pareto frontier
 ◆ SAC-Lagrangian (penalty-based)
 ■ FDPI (constraint-based)